

## Critical safety rule — simulated results must never become real results

The Admin Scenario Runner must be designed so simulated/fake results can never be confused with official real results.

This is a hard safety requirement.

The app must not allow a situation where fake results entered before the tournament remain in the database and later cause users to earn real points, change real leaderboards, affect real bracket health, or appear as official tournament results.

Simulated results are for staging/testing only.

They must not pollute the real result source of truth.

## Required separation

The system must clearly separate:

| Result type                  | Purpose                             | Can affect real user scoring?                       |
|------------------------------|-------------------------------------|-----------------------------------------------------|
| Official results             | Real tournament operation           | Yes                                                 |
| Simulated results            | Admin/testing scenario only         | No, unless the app is explicitly in simulation mode |
| Preview-only scenario output | Calculation preview without writing | No                                                  |

Official results and simulated results must not share an ambiguous state.

A fake score must never look like a real score to the scoring engine, leaderboard engine, Bracket Health, Prediction Trends or public Results page unless the whole app is intentionally running in a clearly marked simulation/test mode.

## Preferred safety model

The safest model is one of these:

### Option A — preview-only simulation

The Scenario Runner calculates simulated tournament states without writing results into the official results table.

This is safest, but may not test the full app as deeply.

## Option B — separate simulation namespace

Simulated results may be stored, but only in a clearly separate simulation namespace/source that cannot be read by normal scoring unless simulation mode is active.

Possible concepts:

```
source = simulation
scenario_id = default_scenario
simulation_run_id = abc123
is_simulated = true
```

The exact implementation should follow the current repo structure, but the rule is non-negotiable:

Normal product scoring must ignore simulated results.

## Option C — local/staging-only seeded scenario

Simulated results can be applied only in local or Euro staging, behind strict environment checks, and must be cleared before any real tournament operation.

This is useful, but it still needs source separation so fake data cannot be accidentally treated as official.

## Real scoring protection

The real scoring engine must only score from official result data.

It must ignore:

- simulated results;
- preview-only results;
- scenario-runner results;
- test fixtures;
- fake clock-generated scores;
- stale simulation data;
- any result marked `source = simulation` or equivalent.

If the app cannot guarantee this, the Scenario Runner should remain preview-only.

## Fake clock protection

Changing the fake clock must not automatically make fake results official.

The fake clock controls the perceived date/time for testing.

It must not by itself:

- create official results;
- award real points;
- mark real matches complete;
- publish real standings;
- change production-visible bracket health.

If simulated scores are generated based on fake clock time, they must remain inside simulation mode or simulation source.

## Pre-tournament safety

This is especially important before the tournament starts.

The system must prevent this failure case:

```
Admin applies fake results before the tournament for testing.  
Fake results remain in the database.  
The real tournament starts.  
Users are awarded points from the fake results.  
Leaderboards become wrong.
```

This must be impossible.

Required protections:

- simulated results clearly marked as simulated;
- normal scoring ignores simulated results;
- clear/reset simulation control;
- warning if simulated results exist;
- admin visible simulation status;
- deployment/pre-launch check that fails or warns if simulated results are present in a live/official mode;
- no simulated result can become official without an explicit, separate official result-entry action.

## Simulation mode indicator

If simulated results are active, Admin should show a clear warning:

```
Simulation mode active
Fake results are being used for testing only.
Official scoring is not using these results.
```

If any public/staging user-facing page is allowed to read simulation state for testing, it should be visually obvious in staging/dev that the app is in simulation mode.

Do not show simulation banners in production unless production simulation is explicitly impossible and hidden from public users.

## Reset and cleanup controls

Admin Scenario Runner must include a safe reset.

Required controls:

```
Clear simulated results
Reset scenario state
Return to clean pre-results state
```

Clearing simulated results should remove or deactivate all scenario-generated data without affecting:

- user predictions;
- official results;
- official scoring;
- league membership;
- accounts;
- real configuration.

## Official result entry remains separate

The proper official result-entry section remains the only way to create official result data.

The Scenario Runner must not be a shortcut into official results.

It should not silently write fake scores into the same state that official scoring reads.

If an admin wants to enter real results, they must use the official result-entry process, not the Scenario Runner.

## Environment guardrails

The Scenario Runner must fail closed outside approved environments.

At minimum:

- allowed in local development;
- allowed in Euro staging if explicitly enabled;
- blocked for WC26 production;
- blocked for any live/official production environment unless a future explicit admin-only simulation design is approved.

The feature must include strong project/environment checks before writing or activating simulated results.

## **Audit and visibility**

Admin should be able to see:

- whether simulation mode is active;
- which scenario is active;
- who applied it;
- when it was applied;
- fake clock time used;
- checkpoint/match simulated up to;
- whether simulated results have been cleared.

This audit trail is admin-only.

## **Acceptance criteria — simulation safety**

The Scenario Runner is not acceptable unless all of these are true:

- simulated results cannot be mistaken for official results;
- normal scoring ignores simulated results;
- fake clock does not create official results;
- simulated results cannot award real points unless the app is explicitly in simulation/test mode;
- simulated results can be cleared safely;
- user predictions are not changed by simulation;
- official results are not changed by simulation;
- real leaderboards are not polluted by fake scores;
- Bracket Health does not use fake scores in normal mode;
- Prediction Trends does not use fake scores in normal mode;
- pre-launch checks warn or fail if simulated results are present in a live/official mode;
- WC26 production remains untouched and fail-closed.

## **Summary rule**

Use this rule throughout implementation:

Fake clock + fake scores can simulate the tournament for testing, but they must never become the official result source of truth.